# Supplement to: When Does Score Diffusion Help Graph Conformal Prediction?

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This supplement contains the full proofs and secondary theory behind the main paper, followed by supporting diagnostics from the same stored experiments. References prefixed by "main paper" refer to the separately compiled manuscript.

## S1 Proofs of the main results

We use the local contextual stochastic block model (CSBM) idealisation and score assumptions in main-paper Definition 1 and Assumption 1. All moments condition on the focal label and degree $d \geq 1$, with $K \geq 2$, $\tau > 0$ and $\Delta > 0$. Write $a = (Kh-1)/(K-1)$ for population adjusted homophily. Uniform population endpoint classes give chance agreement $1/K$. Finite graphs have random endpoint proportions and edge agreement. Their observed $h_{\mathrm{adj}}$ need not equal $a$, even with balanced node counts. Our simulations draw Poisson edges, remove self-loops and deduplicate edges. Their finite neighbourhoods only approximate the independent local model.

### S1.1 Neighbour signal (main-paper Lemma 1)

*Proof.* A neighbour $w$ has $y_w = y_v$ with probability $h$, giving $\mathbb{E}[s(w, y_v)] = \mu + (1-h)\Delta$. For $k \neq y_v$, its label is $k$ with probability $(1-h)/(K-1)$, hence $\mathbb{E}[s(w,k)] = \mu + \Delta\big(1 - \frac{1-h}{K-1}\big)$. Averaging preserves these means. Their difference is $\Delta\big(h - \frac{1-h}{K-1}\big) = \Delta a$. □

### S1.2 Separation and optimal weight (main-paper Proposition 1)

*Proof.* Main-paper Lemma 1 gives $\Delta_\lambda = \Delta(1 - \lambda + \lambda a)$. Total variance gives neighbour-score variance $\tau^2 + h(1-h)\Delta^2$. The independent neighbour draws and score residuals imply $\mathrm{Var}(\bar{s}) = \tau^2\rho^2/d$ and zero own/neighbour covariance, yielding main-paper Eq. (5). Put $c = \rho^2/d > 0$ and $Q(\lambda) = (1-\lambda)^2 + c\lambda^2$. For $\Delta_\lambda > 0$ and $\lambda > 0$, squaring the separation-to-noise ratio (SNR) comparison gives

$$(1 - \lambda + \lambda a)^2 - Q(\lambda) = \lambda\{2a - \lambda(2a - a^2 + c)\} > 0,$$

which is main-paper Eq. (6). A non-positive gap cannot improve on $r_0 = \Delta/\tau > 0$. Over the full interval $[0, 1]$,

$$\frac{d}{d\lambda}\frac{r_\lambda}{r_0} = \frac{a - \lambda(a + c)}{Q(\lambda)^{3/2}}.$$

For $a > 0$, the derivative changes from positive to negative at $a/(a+c)$. For $a \leq 0$, its numerator $a(1 - \lambda) - c\lambda$ is negative on $(0, 1]$. Thus the unique maximiser is $\lambda^\star = a^+/(a^+ + c)$. □

### S1.3 Non-positive homophily (main-paper Corollary 1)

*Proof.* For $a \leq 0$, the derivative above is negative on $(0, 1]$, giving $r_\lambda < r_0$ for every positive weight and finite $d \geq 1$. This implies the stated non-strict inequality. □

### S1.4 Valid selectors (main-paper Proposition 2)

*Proof.* Condition on the graph, features, fitted classifier, all labels (including pool labels), and all auxiliary randomness, including the adaptive prediction sets (APS) draws. This fixes the finite transformed score table and its true-label entries $t_v = \tilde{S}_{v,y_v}$ before the final split. Conditioning on hidden labels here gives the algorithm no access to them.

Draw a uniform permutation $\Pi$ of the $N$ pool nodes. Use its first $m$ nodes for calibration and $u = \Pi_{m+1}$ as the representative test node. The sequence $(t_{\Pi_1}, \ldots, t_{\Pi_m}, t_u)$ is exchangeable sampling without replacement from fixed values. Independent tie-breaking for the rank analysis makes the rank of $t_u$ uniform on $\{1, \ldots, m + 1\}$. Let $j = \lceil (m + 1)(1 - \alpha) \rceil$. For $j \leq m$, every rank at most $j$ is accepted by the calibration $j$-th order statistic. The non-strict score comparison also accepts ties, hence

$$\mathbb{P}\{y_u \in \mathcal{C}(u) \mid \text{conditioned inputs}\} \geq j/(m + 1) \geq 1 - \alpha.$$

If $j = m + 1$, $\hat{q} = +\infty$ and coverage is one. Integrating over the conditioned inputs proves the marginal bound. After holding out a tuning subset, the same argument applies to the remaining pool, conditional on the subset and its labels. The transformation and its randomness must be independent of the final calibration/test assignment. □

## S2 Degree threshold for efficiency

**Corollary S1 (Degree and score efficiency).** *For $a > 0$, $\lambda^\star$ increases with $d$ and tends to one. At fixed $\lambda \in (0, 1]$, main-paper Eq. (6) holds exactly when $d > d^\dagger = \rho^2/[a(a + 2/\lambda - 2)]$. A global weight can therefore over-diffuse nodes with $d \leq d^\dagger$.*

Holding $h, \Delta, \tau$ fixed, equality at the cutoff gives no improvement. If $d^\dagger < 1$, all non-isolated degrees satisfy the condition.

*Proof.* For $a > 0$ and $0 < \lambda \leq 1$, $a + 2/\lambda - 2 > 0$. Rearranging main-paper Eq. (6) gives $a(a + 2/\lambda - 2) > \rho^2/d$, hence the cutoff. Also $\lambda^\star = ad/(ad + \rho^2)$ has derivative $a\rho^2/(ad + \rho^2)^2 > 0$ and limit one. □

## S3 From separation to Gaussian set size

Fix degree $d \geq 1$ and write $p_{\text{wrong}} = (1 - h)/(K - 1)$ for the probability that a neighbour carries a particular wrong label $k \neq y_v$. Total variance gives

$$\tau^2_{\lambda,k} = \tau^2 \left[ (1 - \lambda)^2 + \frac{\lambda^2 \rho_k^2}{d} \right], \qquad \rho_k^2 = 1 + p_{\text{wrong}}(1 - p_{\text{wrong}}) \frac{\Delta^2}{\tau^2}. \tag{S1}$$

Approximate the diffused true and wrong scores by Gaussians with means $\mu_t = \mu + \lambda(1 - h)\Delta$ and $\mu_t + \Delta_\lambda$ and variances $\tau^2_\lambda$ and $\tau^2_{\lambda,k}$. For a homogeneous fixed-degree population or degree-specific population calibration, the true-score quantile is $q_d = \mu_t + z_{1-\alpha}\tau_\lambda(d)$ and

$$\mathbb{E}[|\mathcal{C}(v)| \mid d] \approx 1 - \alpha + (K - 1)\Phi\left( \frac{z_{1-\alpha}\tau_\lambda(d) - \Delta_\lambda}{\tau_{\lambda,k}(d)} \right). \tag{S2}$$

Here $\Phi$ is the standard normal cumulative distribution function and $z_{1-\alpha}$ its quantile. Linearity of expectation suffices, without independent candidate-label inclusion. For $K = 2$, the variances coincide and expected size decreases with the separation-to-noise ratio (SNR) $r_\lambda$. For $K > 2$, variances generally differ, preventing an SNR-only ordering.

Our experiments pool calibration nodes across degrees. Under the Gaussian approximation, let $\omega_d$ be the population share at degree $d$ and fix $\lambda$. The shared threshold $q$ instead solves

$$\sum_d \omega_d \Phi\left( \frac{q - \mu_t(d)}{\tau_\lambda(d)} \right) = 1 - \alpha. \tag{S3}$$

Expected size is the degree-weighted sum of true- and wrong-label inclusion probabilities evaluated at this shared $q$. Averaging the degree-specific formula need not give that size. At $d = 0$, use original-score mean $\mu$, variance $\tau^2$ and gap $\Delta$. The crossover heuristic in main-paper Sect. 7.1 averages inverse degree in the SNR condition, without solving this mixture calibration problem. Bounded APS scores need not be Gaussian, and finite calibration adds quantile error. These formulas provide intuition within the stated approximation.

## S4 Correlated-score moment extension

Retain the conditional means and marginal variances of the main-paper score model. Let distinct neighbour scores for the true candidate label have correlation

$\rho_{\mathrm{nn}}$, and each node-to-neighbour pair have correlation $\kappa$, conditional on the focal label and degree. Own variance is $\tau^2$, neighbour variance $\tau^2\rho^2$, and node-to-neighbour covariance $\kappa\tau^2\rho$. For a feasible covariance matrix,

$$\mathrm{Var}(\bar{s}) = \tau^2\rho^2\{1 + (d-1)\rho_{\mathrm{nn}}\}/d, \tag{S4}$$

$$\mathrm{Cov}(s, \bar{s}) = \kappa\tau^2\rho, \tag{S5}$$

$$\mathrm{Var}(\tilde{s}) = \tau^2\left[(1-\lambda)^2 + \lambda^2\rho^2\frac{1 + (d-1)\rho_{\mathrm{nn}}}{d} + 2\lambda(1-\lambda)\kappa\rho\right]. \tag{S6}$$

For $d > 1$, feasibility requires $-1/(d-1) \le \rho_{\mathrm{nn}} \le 1$ and $\kappa^2 \le \{1+(d-1)\rho_{\mathrm{nn}}\}/d$. For $d = 1$, $|\kappa| \le 1$. Setting $\rho_{\mathrm{nn}} = \kappa = 0$ recovers main-paper Eq. (5). Positive $\rho_{\mathrm{nn}}$ leaves a noise floor as degree grows. The optimum under correlated scores need not approach one. We have not estimated these moments for our graph neural networks (GNNs). This extension identifies a limitation of the independent-score rule.

## S5 Supplementary experimental details

We compare APS, diffusion adaptive prediction sets (DAPS), and our SNR-based selector (SNR-$\lambda$), using a graph convolutional network (GCN).

**Table S1.** Coverage at nominal 0.95 (GCN, 50 runs). Weighted and Mondrian columns apply to SNR-$\lambda$. Uniform rows are reference splits. Arrows identify the calibration and test populations.

| | | Uncorrected | | | SNR-$\lambda$ corrected | |
|---|---|---|---|---|---|---|
| Graph | Calibration → test | APS | DAPS | SNR-$\lambda$ | weighted | Mondrian |
| Computers | low→high | .969 | .984 | .985 | .954 | .967 |
| Computers | high→low | .956 | .941 | .942 | .957 | .964 |
| Computers | uniform (reference) | .949 | .950 | .950 | .951 | .955 |
| Amazon-ratings | low→high | .951 | .954 | .954 | .952 | .956 |
| Amazon-ratings | high→low | .951 | .945 | .944 | .950 | .952 |
| Amazon-ratings | uniform (reference) | .949 | .949 | .950 | .950 | .953 |
| ogbn-arxiv | $\le$2017 → $\ge$2019 | .969 | .958 | .950 | .949 | .953 |
| ogbn-arxiv | uniform (reference) | .949 | .949 | .949 | .950 | .953 |

Table S1 completes the comparison in main-paper Sect. 7.4, including reverse degree shifts and Mondrian calibration. On Computers, reversing the shift changes SNR-$\lambda$ coverage from 0.942 to 0.985. Mondrian calibration is conservative in these runs. The table allows direct comparison with propensity weighting for each shift direction.

*Original-protocol checks.* The DAPS check in main-paper Sect. 7.2 uses twenty labels per class and dataset-specific adaptive targets: $\alpha = 0.060$ on Cora-ML,

0.055 on CiteSeer (CitationFull), 0.070 on PubMed, and 0.067 on Computers. Mean set-size changes from APS are $-0.53$, $-0.39$, $-0.05$ and $-0.78$ labels, respectively, compared with $-0.50$, $-0.40$, $+0.02$ and $-0.77$ labels in the original DAPS Table 1. The separate official-code conformalized graph neural network (CF-GNN) diagnostic uses three runs at $\alpha = 0.1$. For the main comparison, the artifact documents the 500 canonical CF-GNN rows in `docs/cfgnn-selection.csv`. Files are sorted lexicographically and the first row per dataset/model/seed/split/alpha key is retained. Unsuffixed files therefore precede overlapping `_rev` rows, while later files supply missing keys. Alternative copies are not averaged.

## S6 Agreement-based selector specifications

The following rules supplement the comparison in main-paper Sect. 5. Here $\mathrm{clip}(x, 0, 1)$ truncates $x$ to $[0, 1]$.

*Label-free, node-wise.* Heterophily-aware diffused conformal prediction (HEAD-CP) (main-paper Sect. 5) sets a node- or edge-wise coefficient from predicted labels or probabilities. We re-implement its three variants with fixed published constants, plus our original rule: $\lambda_v = \mathrm{clip}(2g_v - 1, 0, 1)$ with soft agreement $g_v = d_v^{-1} \sum_{w \in \mathcal{N}(v)} \langle \pi(v), \pi(w) \rangle$.

*Label-free, graph-level.* We average soft agreement over edges and map it to $\lambda = 0.75\,\mathrm{clip}(\cdot, 0, 1)$ under two normalisations: the analogue of main-paper Eq. (4) with predicted class shares, and a self-normalised version that divides by the mean self-agreement $\langle \pi(v), \pi(v) \rangle$ minus chance. The constant 0.75 was fixed on Cora and Roman-empire before other graphs were evaluated.

*Label-based gate.* The same map applied to the adjusted homophily $\hat{a}$ of training edges gives a label gate. We adapt the gate of homophily-gated DAPS (HG-DAPS, main-paper Sect. 5) to training edges. It computes training-edge adjusted homophily $\hat{a}$ and sets $\lambda = 0.5\,\mathrm{clip}\big(\hat{a}\, m_{\mathrm{tr}}/(m_{\mathrm{tr}} + 100), 0, 1\big)$, with $m_{\mathrm{tr}}$ the number of such edges. We retain ordinary APS scores on training nodes. The original method instead uses one-hot scores on its labelled context, which covers half the nodes.

## S7 Pseudocode and implementation notes

Algorithm S1 gives the full prediction pipeline for the model-derived SNR-$\lambda$ selector. The classifier is already fitted. $V_{\text{val}}$ is its held-out validation set, and $V_{\text{pool}}$ contains the remaining nodes. These sets and $\mathcal{V}_{\text{tr}}$ are disjoint. Assume $K \geq 2$, a nonempty validation set, positive validation true-score variance, and defined training-edge homophily estimates.

**Algorithm S1:** SNR-based score diffusion and split conformal prediction

**Input:** Graph $\mathcal{G}$, fitted probabilities $\pi(v)$, training and validation labels, pool $V_{\text{pool}}$, calibration size $m$, error rate $\alpha$.
**Output:** Prediction sets $\mathcal{C}(v)$ for $v \in \mathcal{V}_{\text{test}}$.

Draw independent $U_v \sim \text{Unif}[0,1]$ for all nodes
Compute APS scores $S_{vk}$ using main-paper Eq. (2)
Estimate edge homophily $\hat{h}$ and adjusted homophily $\hat{a}$ on edges with both endpoints in $\mathcal{V}_{\text{tr}}$
$S_{\text{true}} \leftarrow \{S_{v,y_v} : v \in V_{\text{val}}\}$
$S_{\text{wrong}} \leftarrow \{S_{v,k} : v \in V_{\text{val}},\ k \neq y_v\}$
$\hat{\Delta} \leftarrow \text{mean}(S_{\text{wrong}}) - \text{mean}(S_{\text{true}})$, $\hat{\tau}^2 \leftarrow \text{mean}[(S_{\text{true}} - \text{mean}(S_{\text{true}}))^2]$
$\hat{\rho}^2 \leftarrow 1 + \hat{h}(1-\hat{h})\hat{\Delta}^2/\hat{\tau}^2$, $\hat{a}^+ \leftarrow \max(\hat{a}, 0)$
**foreach** *node* $v$ **do**
  **if** $d_v = 0$ **then**
    $\lambda_v \leftarrow 0$, $\tilde{S}_{v,:} \leftarrow S_{v,:}$
  **else**
    $\lambda_v \leftarrow \hat{a}^+/(\hat{a}^+ + \hat{\rho}^2/d_v + 10^{-12})$
    $M_{v,:} \leftarrow d_v^{-1} \sum_{w \in \mathcal{N}(v)} S_{w,:}$
    $\tilde{S}_{v,:} \leftarrow (1-\lambda_v)S_{v,:} + \lambda_v M_{v,:}$
Draw $\mathcal{V}_{\text{cal}} \subset V_{\text{pool}}$ uniformly with $|\mathcal{V}_{\text{cal}}| = m$, $\mathcal{V}_{\text{test}} \leftarrow V_{\text{pool}} \setminus \mathcal{V}_{\text{cal}}$
Reveal calibration labels, set $j \leftarrow \lceil (m+1)(1-\alpha) \rceil$
**if** $j > m$ **then**
  $\hat{q} \leftarrow +\infty$
**else**
  $\hat{q} \leftarrow j$-th smallest value in $\{\tilde{S}_{v,y_v} : v \in \mathcal{V}_{\text{cal}}\}$
**return** $\mathcal{C}(v) = \{k : \tilde{S}_{vk} \leq \hat{q}\}$ *for all* $v \in \mathcal{V}_{\text{test}}$

*Implementation details.* The score definitions, diffusion rule and validity argument are in main-paper Sects. 3 to 5. Algorithm S1 specifies their execution order and the empirical estimators. Training-edge class shares use degrees within the training-induced subgraph. In `src/scores.py`, APS is implemented as the mass preceding the candidate plus $\xi_v \pi_k(v)$. This equals the manuscript's formula with $U_v = 1 - \xi_v$. The same random draws are reused across methods in each run. The variance uses divisor $|V_{\text{val}}|$, and the $10^{-12}$ term matches `snr_lambda`. The stated preconditions exclude undefined homophily or zero-variance estimates. The denominator offset alone does not resolve those cases. `src/conformal.py` implements the order statistic and the $+\infty$ convention.

*Computational cost.* With $n$ nodes, $K$ classes and $|\mathcal{E}|$ edges, sorting for APS costs $O(nK \log K)$ and diffusion costs $O(|\mathcal{E}|K)$. Scores require $O(nK)$ storage, in addition to the sparse graph. The sparse multiplication computes all neighbour means together. The node loop in Algorithm S1 makes the operations explicit.

## S8 Evaluation metrics

The formulas below specify the metrics summarised in main-paper Sect. 6. For one run, let $T = \mathcal{V}_{\text{test}}$, $N = |T|$, $c_v = \mathbb{1}\{y_v \in \mathcal{C}(v)\}$ and $\ell_v = |\mathcal{C}(v)|$. Here $\mathbb{1}\{\cdot\}$ equals one when its condition holds and zero otherwise. Metrics are computed per run before aggregation.

*Coverage and set size.*

$$\widehat{\text{Cov}} = \frac{1}{N}\sum_{v\in T} c_v, \qquad L = \frac{1}{N}\sum_{v\in T} \ell_v. \tag{S7}$$

Coverage estimates how often the true label is retained. Its target is $1-\alpha$. Mean set size $L$ measures the remaining ambiguity: smaller is better when coverage is comparable. For a method $M$ relative to APS,

$$\text{Change}(M) = 100\left(\frac{\bar{L}_M}{\bar{L}_{\text{APS}}} - 1\right)\%, \tag{S8}$$

where bars denote run means for the specified graph and classifier. Positive values indicate inflation. Negative values indicate shrinkage. This ratio of mean sizes differs from the mean of run-wise ratios used for regret below.

*Coverage within groups.* For a group $G \subseteq T$, define

$$\widehat{\text{Cov}}(G) = \frac{\sum_{v\in G} c_v}{|G|}, \quad \text{Cov}_{\min} = \min_{G\in\mathcal{B}} \widehat{\text{Cov}}(G), \quad D_{\max} = \max_{G\in\mathcal{B}} |\widehat{\text{Cov}}(G) - (1-\alpha)|. \tag{S9}$$

For worst-class coverage, $\mathcal{B}$ groups nodes by true label. For worst-degree coverage, it contains ten approximately equal-sized groups formed by sorting test-node degrees, with random tie-breaking. The implementation retains groups with at least 20 test nodes. It returns a missing value if none qualify. Thus the reported worst group is the worst among eligible groups. $D_{\max}$, also stored in the artifact, measures absolute deviation, counting both undercoverage and overcoverage. Reported worst-group coverage averages per-run minima. It is not the minimum after pooling runs.

*Worst-slice coverage (WSC).* A slab selects nodes whose feature embedding $z_v$ projects into an interval:

$$G(\ell, b, \theta) = \{v : \ell \le \theta^\top z_v \le b\}, \qquad \|\theta\|_2 = 1. \tag{S10}$$

Randomly divide $T$ into a search half $T_A$ and an evaluation half $T_B$. Among candidate slabs containing at least 10% of $T_A$, select

$$G^* \in \arg\min_G \widehat{\mathrm{Cov}}(G \cap T_A), \qquad \widehat{\mathrm{WSC}} = \widehat{\mathrm{Cov}}(G^* \cap T_B). \tag{S11}$$

The experiments search 200 random directions and endpoints on a 20-interval projection-quantile grid. The minimum-size constraint applies to the search half. The held-out slab may contain fewer nodes, and an empty slab gives a missing value. This finite search diagnoses feature-defined subgroups without reporting their optimistically selected search coverage. It is neither an exhaustive worst-subgroup guarantee nor a substitute for class-level coverage.

*Oracle regret and aggregation.* For cell $c$ (one graph and classifier pair), run $r$ and selector $M$, let $L_{c,r,M}$ be mean set size. The retrospective scalar-grid reference is

$$L^*_{c,r} = \min_{\lambda \in \{0, 0.1, \ldots, 1\}} L_{c,r,\lambda}, \qquad R_{c,r,M} = 100 \left( \frac{L_{c,r,M}}{L^*_{c,r}} - 1 \right). \tag{S12}$$

Regret is expressed in percent. With $J_c$ paired runs in cell $c$, set $\bar{R}_{c,M} = J_c^{-1} \sum_r R_{c,r,M}$. Over the $C$ evaluated cells,

$$R_{\mathrm{mean},M} = \frac{1}{C} \sum_c \bar{R}_{c,M}, \; R_{\mathrm{worst},M} = \max_c \bar{R}_{c,M}, \; P_{2,M} = \frac{100}{C} \sum_c \mathbb{1}\{\bar{R}_{c,M} \le 2\}. \tag{S13}$$

Cells have equal weight. Homophilous and heterophilous means restrict the same average to their respective cells. "Worst" means the worst cell mean, not the worst run. $P_{2,M}$ is the percentage of cells within 2% of the reference. Negative regret is possible because node-wise or signed rules lie outside the oracle's non-negative scalar grid, or a method calibrates on a different subset (as DAPS-cal does). The random-grid baseline uses $11^{-1} \sum_\lambda L_{c,r,\lambda}$, its exact expected size under a uniform grid choice, before computing regret.

*Additional stored diagnostics.* Singleton rate is $N^{-1} \sum_v \mathbb{1}\{\ell_v = 1\}$. Empty-set rate is $N^{-1} \sum_v \mathbb{1}\{\ell_v = 0\}$. Size-stratified coverage violation is $\max_B |\widehat{\mathrm{Cov}}(B) - (1-\alpha)|$, using nonempty-size bins $\{1\}$, $\{2,3\}$, $\{4,5,6\}$, $\{7,\ldots,10\}$, $\{11,\ldots,20\}$, $\{21,\ldots,50\}$ and $\{51,\ldots,K\}$, restricted to feasible sizes and bins with at least 20 nodes. The implementation returns zero when no bin qualifies. This convention is not evidence of perfect conditional coverage. Classifier accuracy is $N^{-1} \sum_v \mathbb{1}\{\arg\max_k \pi_k(v) = y_v\}$, which measures single-label prediction. For weights $w_i$, effective sample size (ESS) is

$$\mathrm{ESS} = \frac{(\sum_i w_i)^2}{\sum_i w_i^2}. \tag{S14}$$

Small values indicate that a few nodes dominate the weighted calibration sample.

*Uncertainty summaries.* For a reported mean $\bar{x}$, each bootstrap replicate resamples the relevant units with replacement and recomputes the mean. The 2.5th and 97.5th percentiles of 1000 replicates define a 95% percentile interval. Tables showing $\bar{x} \pm b_{\mathrm{CI}}$ use half its width, $b_{\mathrm{CI}} = (q_{0.975} - q_{0.025})/2$. Runs are the units for run-level summaries. Cells are the units for the regret-mean interval, which is descriptive because classifiers share graphs. Crossover intervals instead resample graph seeds, as described in main-paper Sect. 7.1.